Task 1: Evaluate RNA secondary structure prediction using LILP model

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1 Run LILP prediction for an RNA sequence

I ran the LILP prediction for the following RNA sequence. The dot-bracket notation below represents the supplied reference structure.

```
>Azoa.spec._AM406670
GGCGGGCCUCCCCGCAUUGCAAAGUAGCCAACCUGGUCAGGUCCGGAAGGAAGCAGCCACAACUAUUGAAGCAAGUGCCGGGGGUUGGGCUCGCC
((((((((((((((.((((..(((((......((((....(((....)))....))))...))))))...))))))...)))))))..)).)))))))
```

2 Compare energies of LILP generated structure with reference and DP structures.

2.1 Calculate energies using RNAeval for all structures for a selected sequence

The following listing presents the energy contributions reported by RNAeval for the supplied structure.

```
External loop                           :     0
Interior loop (  1, 95) GC; (  2, 94) GC:  -330
Interior loop (  2, 94) GC; (  3, 93) CG:  -340
Interior loop (  3, 93) CG; (  4, 92) GC:  -240
Interior loop (  4, 92) GC; (  5, 91) GU:  -150
Interior loop (  5, 91) GU; (  6, 90) GC:  -210
Interior loop (  6, 90) GC; (  7, 88) CG:    40
Interior loop (  7, 88) CG; (  8, 87) CG:  -330
Interior loop (  8, 87) CG; (  9, 84) UG:   330
Interior loop (  9, 84) UG; ( 10, 83) CG:  -150
Interior loop ( 10, 83) CG; ( 11, 82) CG:  -330
Interior loop ( 11, 82) CG; ( 12, 81) CG:  -330
Interior loop ( 12, 81) CG; ( 13, 80) CG:  -330
Interior loop ( 13, 80) CG; ( 14, 79) GC:  -240
Interior loop ( 14, 79) GC; ( 15, 75) CG:   320
Interior loop ( 15, 75) CG; ( 17, 74) UA:   170
Interior loop ( 17, 74) UA; ( 18, 73) UA:   -90
Interior loop ( 18, 73) UA; ( 19, 72) GC:  -210
Interior loop ( 19, 72) GC; ( 20, 71) CG:  -340
Interior loop ( 20, 71) CG; ( 23, 67) AU:   220
Interior loop ( 23, 67) AU; ( 24, 66) GU:   -60
Interior loop ( 24, 66) GU; ( 25, 65) UA:  -140
Interior loop ( 25, 65) UA; ( 26, 64) AU:  -130
Interior loop ( 26, 64) AU; ( 27, 63) GC:  -210
Interior loop ( 27, 63) GC; ( 34, 59) UA:   490
Interior loop ( 34, 59) UA; ( 35, 58) GC:  -210
Interior loop ( 35, 58) GC; ( 36, 57) GC:  -330
Interior loop ( 36, 57) GC; ( 37, 56) UG:  -250
Interior loop ( 37, 56) UG; ( 42, 51) UA:   290
Interior loop ( 42, 51) UA; ( 43, 50) CG:  -240
Interior loop ( 43, 50) CG; ( 44, 49) CG:  -330
Hairpin  loop ( 44, 49) CG              :   330
```

The evaluated structure is shown below. The reported total energy is $E_{\mathrm{RNAeval}} = -33.30$.

```
((((((((((((((.((((..(((((......((((....(((....)))....))))...))))))...))))))...)))))))..)).)))))))
```

2.2 Calculate energies for all structures for a selected sequence according to the LILP model.

The following listing presents the energy contributions obtained according to the LILP model.

```
External loop                           :     0
Interior loop (  1, 95) GC; (  2, 94) GC:  -326
Interior loop (  2, 94) GC; (  3, 93) CG:  -342
Interior loop (  3, 93) CG; (  4, 92) GC:  -236
Interior loop (  4, 92) GC; (  5, 91) GU:  -153
Interior loop (  5, 91) GU; (  6, 90) GC:  -211
Interior loop (  6, 90) GC; (  7, 89) CG:  -342
Interior loop (  7, 89) CG; (  8, 88) CG:  -326
Interior loop (  8, 88) CG; (  9, 87) UG:  -211
Interior loop (  9, 87) UG; ( 10, 83) CG:   319
Interior loop ( 10, 83) CG; ( 11, 82) CG:  -326
Interior loop ( 11, 82) CG; ( 12, 81) CG:  -326
Interior loop ( 12, 81) CG; ( 13, 80) CG:  -326
Interior loop ( 13, 80) CG; ( 14, 78) GC:   143
Interior loop ( 14, 78) GC; ( 15, 77) CG:  -342
Interior loop ( 15, 77) CG; ( 16, 76) AU:  -211
Interior loop ( 16, 76) AU; ( 17, 74) UA:   269
Interior loop ( 17, 74) UA; ( 18, 73) UA:   -93
Interior loop ( 18, 73) UA; ( 19, 72) GC:  -211
Interior loop ( 19, 72) GC; ( 20, 71) CG:  -342
Interior loop ( 20, 71) CG; ( 23, 67) AU:   150
Interior loop ( 23, 67) AU; ( 24, 66) GU:   -55
Interior loop ( 24, 66) GU; ( 25, 65) UA:  -136
Interior loop ( 25, 65) UA; ( 26, 64) AU:  -133
Interior loop ( 26, 64) AU; ( 27, 63) GC:  -208
Interior loop ( 27, 63) GC; ( 34, 59) UA:   420
Interior loop ( 34, 59) UA; ( 35, 58) GC:  -211
Interior loop ( 35, 58) GC; ( 36, 57) GC:  -326
Interior loop ( 36, 57) GC; ( 37, 56) UG:  -251
Interior loop ( 37, 56) UG; ( 42, 51) UA:   130
Interior loop ( 42, 51) UA; ( 43, 50) CG:  -235
Interior loop ( 43, 50) CG; ( 44, 49) CG:  -326
Hairpin  loop ( 44, 49) CG              :   240
```

The predicted structure and its reported total energy are:

```
GGCGGGCCUCCCCGCAUUGCAAAGUAGCCAACCUGGUCAGGUCCGGAAGGAAGCAGCCACAACUAUUGAAGCAAGUGCCGGGGGUUGGGCUCGCC
((((((((((((((((((((..(((((......((((....(((....)))....))))...))))))...))))).))))...))))))))))
```

$$E_{\mathrm{LILP}} = -45.34.$$

3 Compare predictions by metrics: PPV, STY and INF.

Table 1 compares the prediction metrics with respect to the reference structure. TP, FP, and FN denote true positives, false positives, and false negatives, respectively.

Table 1: Comparison of prediction metrics.

Method	TP	FP	FN	PPV	STY	INF
LILP	26	6	5	0.8125	0.8387	0.8255
RNAeval	25	7	6	0.7813	0.8065	0.7938

4 Draw 2D diagrams for all secondary structures

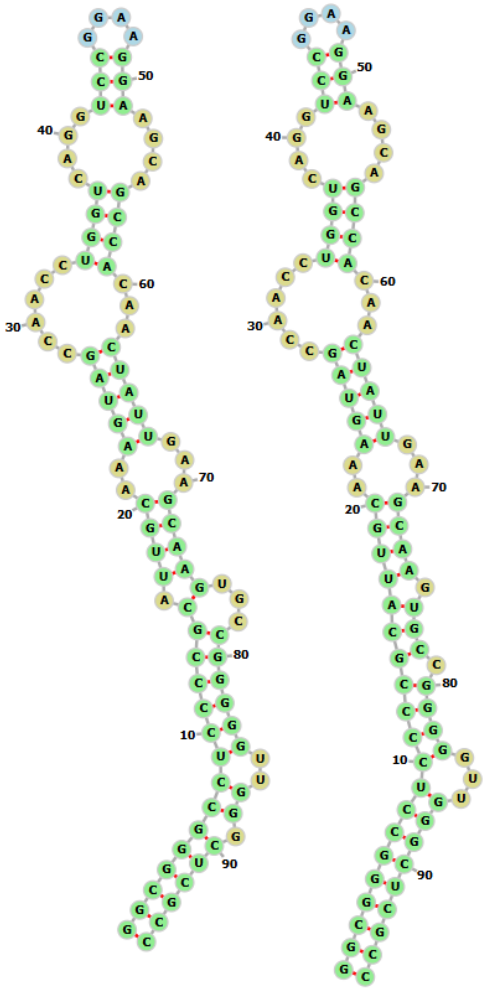


Figure 1: Comparison of the structure labelled RNAeval and the LILP prediction.

In region 5 of Figure 1, the upper structure, labelled RNAeval, contains a bulge adjacent to an internal loop, whereas the LILP structure contains two consecutive bulges.

In region 6, the internal loop in the LILP structure contains one additional guanine (G), and the subsequent bulge is absent.

These differences may result from differences in the energy parameters or energy terms used by the two models, which can favor different base-pair arrangements.